Supplementary Material

Class-conditional conformal prediction masks chemical-support undercoverage in blood-brain barrier screening

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All results use the confirmatory split seeds 100–119; pilot seeds are not reported. Tables in this supplement are generated directly from the analysis outputs by `scripts/make_supplement_tables.py`; numbers quoted in the main text are additionally recomputed by an independent program (Section H).

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A Theory

This section collects every formal statement used in the paper, with complete proofs, and separates exact results from design choices (Table S2). None of the statements is a new conformal method; Propositions S2–S3 and Corollary S1 make precise why the class-conditional guarantee leaves a support gradient, why conditioning on support removes it by redistribution, and how much of that change finite-sample conservativeness alone can produce.

Notation. $Y \in \{0, 1\}$ with class 0 = BBB− (the minority); LAC score $s(x, y) = 1 - \hat{p}_y(x)$; the model $\hat{f}$ is fit on TRAIN and selected on VAL. The stratum map b assigns VAL-quantile bins of maximal Tanimoto support $s_{\max}$, so b is fixed before CAL and TEST are used. $w_{c,b} = \mathbb{P}(b(X) = b \mid Y = c)$; $\mathrm{cov}_c = \mathbb{P}(Y \in C(X) \mid Y = c)$ and $\mathrm{cov}_{c,b} = \mathbb{P}(Y \in C(X) \mid Y = c, b(X) = b)$, where probabilities are over CAL and the test point, conditional on TRAIN and VAL. $k_n = \lceil (n + 1)(1 - \alpha) \rceil$. **Assumption E** (main-text Assumption 1): CAL $\cup \{(X_{n+1}, Y_{n+1})\}$ is exchangeable given TRAIN $\cup$ VAL. **Assumption I**: CAL and the

test point are i.i.d. given TRAIN ∪ VAL (implies E). **Assumption C**: given the class (and stratum), scores have a continuous distribution, so ties occur with probability zero. E holds for random splits; I and C are used only where stated, and every guarantee is claimed for random splits only.

A.1 What a class-level guarantee permits

Proposition S1 (Stratum floor; main-text Proposition 1)**.** *If* $\mathrm{cov}_c \geq 1-\alpha$*, then* $\mathrm{cov}_{c,b} \geq \max(0, 1-\alpha/w_{c,b})$ *for every* b *with* $w_{c,b} > 0$*, and the bound is attained by a class-conditionally valid predictor.*

Proof. By total probability, $\mathrm{cov}_c = \sum_{b'} w_{c,b'}\mathrm{cov}_{c,b'}$. Since each $\mathrm{cov}_{c,b'} \leq 1$, $w_{c,b}\mathrm{cov}_{c,b} \geq \mathrm{cov}_c - \sum_{b' \neq b} w_{c,b'} \geq 1-\alpha-(1-w_{c,b})$; divide by $w_{c,b}$, and coverage is non-negative. For attainment with $w_{c,b} \geq \alpha$, include c with probability one when $b(x) \neq b$ and independently with probability $1-\alpha/w_{c,b}$ when $b(x) = b$: then $\mathrm{cov}_{c,b} = 1-\alpha/w_{c,b}$ and $\mathrm{cov}_c = (1-w_{c,b}) + w_{c,b} - \alpha = 1-\alpha$. If $w_{c,b} < \alpha$, exclude c in b: $\mathrm{cov}_{c,b} = 0$ and $\mathrm{cov}_c = 1 - w_{c,b} > 1-\alpha$. □

The floor depends on the measured class share of the stratum, not on $1/B$. The BBB− share of the low-support tertile is 0.28, 0.35, 0.35 and 0.37 (BBBP random, BBBP scaffold, B3DB random, B3DB scaffold), giving floors of 0.64, 0.71, 0.71 and 0.73 at $\alpha = 0.1$. Observed class-conditional low-support coverage, 0.79–0.87, lies far above these floors: the guarantee permits a much larger failure than occurs.

A.2 Why a shared class threshold produces a support gradient

Proposition S1 says a failure is permitted; the next statement says when it must occur and in which direction. Let $F_{c,b}(t) = \mathbb{P}(s(X,c) \leq t \mid Y = c, b(X) = b)$ be the distribution function of class-c scores in stratum b (given TRAIN and VAL), and $\hat{q}_c$ the class-conditional threshold computed from the class-c calibration scores.

Proposition S2 (Coverage under a shared threshold; main-text Proposition 2)**.** *Under Assumptions I and C, class-conditional calibration gives*

$$\mathrm{cov}_{c,b} = \mathbb{E}\big[F_{c,b}(\hat{q}_c)\big] \qquad \text{for every stratum } b.$$

(a) If $F_{c,b}(t) \leq F_{c,b'}(t)$ *for all* t *(scores in* b *stochastically larger), then* $\mathrm{cov}_{c,b} \leq \mathrm{cov}_{c,b'}$*. (b) If the strata are so ordered,* $F_{c,1} \leq F_{c,2} \leq \cdots \leq F_{c,B}$*, then* $\mathrm{cov}_{c,1} \leq \mathrm{cov}_c \leq \mathrm{cov}_{c,B}$*; in particular, whenever the least supported stratum is under-covered, the best supported stratum is over-covered. (c) An ordering of mean scores,* $\mathbb{E}[s \mid c,b] \geq \mathbb{E}[s \mid c,b']$*, does not imply* $\mathrm{cov}_{c,b} \leq \mathrm{cov}_{c,b'}$*.*

Proof. Under I, the test point is independent of CAL given TRAIN and VAL, and $\hat{q}_c$ is a function of CAL. Conditioning on CAL and on $\{Y = c, b(X) = b\}$, the test point is covered iff $s(X,c) \leq \hat{q}_c$, which has probability $F_{c,b}(\hat{q}_c)$; averaging over CAL gives the identity. (a) follows because the integrand is pointwise ordered. (b) follows from (a) and $\mathrm{cov}_c = \sum_b w_{c,b}\mathrm{cov}_{c,b}$, a convex combination. For (c), let $s \mid b \sim \mathrm{Uniform}(0,1)$ and let $s \mid b'$ take the value 0.05 with probability 0.8 and 0.95 with probability 0.2 (smoothed slightly to satisfy C). Stratum b has the larger mean score (0.50 against 0.23), yet at a shared threshold $\hat{q}_c = 0.9$, $F_{c,b}(0.9) = 0.9 > 0.8 = F_{c,b'}(0.9)$, so b is the better covered stratum. □

Link to the measurements. The mechanism analysis (Table S13, Fig. S2) measures the mean true-class probability $1 - \mathbb{E}[s \mid c,b]$ by tertile, a summary of $F_{c,b}$. A larger drop for BBB− than for BBB+ in every BBB cell, together with the observed coverage ordering, is what part (a) predicts when the drop reflects a stochastic shift. Part (c) explains why the same mean drop on Tox21 SR-ARE does not produce the gradient (main text, Section 6.9): the relevant quantity is the mass of each stratum's score distribution below the shared threshold, not its mean. The reliability diagram (Fig. S3) shows the corresponding calibration shift on B3DB: at a fixed predicted probability, low-support compounds are BBB− more often.

A.3 Joint conditioning: what it guarantees and what it costs

Proposition S3 (Cell coverage band)**.** *Fix a partition of strata determined before CAL and TEST are used. Under E and C, for a (class, stratum) cell with* n *calibration points and* $k_n \leq n$*, class × support calibration*

gives $\mathrm{cov}_{c,b} = k_n/(n+1)$ *in expectation over CAL, hence*

$$1-\alpha \;\le\; \mathrm{cov}_{c,b} \;<\; 1-\alpha+\frac{1}{n+1}.$$

If $k_n > n$ *the cell's threshold is infinite, so label* c *is always included for test points in stratum* b *and* $\mathrm{cov}_{c,b} = 1$.

Proof. Within the cell, the n calibration scores and the test score are exchangeable and almost surely distinct, so the rank of the test score among the $n+1$ is uniform on $\{1,\dots,n+1\}$. The test point is covered iff its rank is at most k_n, which has probability $k_n/(n+1)$. Since $(n+1)(1-\alpha) \le k_n < (n+1)(1-\alpha)+1$, the band follows. □

Consequence: redistribution. Under class-conditional calibration, Proposition S2(b) forces the best supported stratum above target whenever the worst is below it, and nothing bounds that over-coverage. Under class × support calibration, Proposition S3 places every cell, in expectation, between $1-\alpha$ and $1-\alpha+1/(n+1)$: low-support coverage is raised to at least $1-\alpha$, and over-coverage above that cap is removed. This is a property of joint conditioning, not a conservation law that holds for every method. In our data high-support BBB− coverage moves from 0.94–0.97 to 0.85–0.92, and its interval lies entirely below 0.90 in 0 of 12 cells, consistent with the band, which is guaranteed only for random splits.

Corollary S1 (How much smaller cells alone can change coverage)**.** *Split class* c *into* B *strata assigned independently of* (X,Y) *given the class, with* $n_1,\dots,n_B$ *calibration points* ($\sum n_r = n$*, each* $k_{n_r} \le n_r$*). Under E and C, the expected class-level coverage is* $\sum_r \pi_r\, k_{n_r}/(n_r+1)$*, with* π_r *the stratum probabilities. Its difference from class-conditional calibration,* $k_n/(n+1)$*, lies strictly between* $-1/(n+1)$ *and* $\max_r 1/(n_r+1)$*. The difference can be negative, because* k_n *is an integer: at* $\alpha = 0.1$*,* $n = 27$ *gives* $26/28$*, whereas three random cells of nine give* $9/10$*. With three equal strata, any increase is below* $1/(n/3+1)$.

Proof. Random assignment makes each random cell an exchangeable sample of the class, so Proposition S3 applies cell by cell. Each cell's expected coverage lies in $[1-\alpha,\, 1-\alpha+1/(n_r+1))$, so their mixture does too, and class-conditional coverage lies in $[1-\alpha,\, 1-\alpha+1/(n+1))$; subtracting gives both bounds. □

Check against the same-cell-size control. For the realized minority calibration sizes, $1/(n/3+1)$ is 0.043–0.044 on BBBP and 0.013–0.014 on B3DB. Randomly assigned strata raise low-support BBB− coverage by 0.010–0.045 on BBBP and 0.008–0.016 on B3DB (Table S9). The corollary bounds the class-level change, not the change on a support subset, so this is a check of scale rather than a bound; the observed changes are of the predicted size and smaller on the dataset with more calibration data, whereas class × Tanimoto cells raise low-support coverage by 0.036–0.100 beyond random strata.

A.4 Count-based merging

Algorithm S1: Count-based merging (SABM) for a test molecule x. Brackets name the data each step may use.

Input: scores s_i, labels y_i, strata b_i of CAL [CAL]; stratum edges [VAL]; model $\hat{f}$ [TRAIN, VAL]; level α; budget n_{req} fixed in advance

for *each candidate label* $c \in \{0,1\}$ **do**

- $N_\beta \leftarrow \#\{i \in \mathrm{CAL} : y_i = c,\, b_i = \beta\}$ for $\beta = 1,\dots,B$; $N_{b(x)} \leftarrow N_{b(x)} + 1$; `// bag counts [CAL, x; no test label]`
- Traverse $\beta = 1,\dots,B$ from low to high support, closing a group when its accumulated count reaches n_{req}; merge a deficient final group into the preceding one ; `// counts only`
- $g \leftarrow$ group containing $b(x)$; $m \leftarrow \#\{i : y_i = c,\, b_i \in g\}$; $\hat{q} \leftarrow$ the $\lceil (m+1)(1-\alpha) \rceil$-th smallest of $\{s_i : y_i = c,\, b_i \in g\}$, or $+\infty$ if $\lceil (m+1)(1-\alpha) \rceil > m$ [CAL];
- include c in $C(x)$ if and only if $s(x,c) \le \hat{q}$ [TEST x only];

Theorem S1 (Class- and group-conditional validity; main-text Theorem 1)**.** *Let Γ_c be the partition computed for candidate label c from the bag of calibration and test points (Algorithm S1, lines 2–3). Under Assumption E, for every class c, partition γ and group $g \in \gamma$ with $\mathbb{P}(Y = c, \Gamma_c = \gamma, b(X) \in g) > 0$, $\mathbb{P}(Y \in C(X) \mid Y = c,\ \Gamma_c = \gamma,\ b(X) \in g) \geq 1 - \alpha$; consequently* $\mathrm{cov}_c \geq 1 - \alpha$.

Proof. Let $\mathcal{B}$ be the multiset of the $n+1$ points CAL $\cup\{(X_{n+1}, Y_{n+1})\}$. *Bag symmetry.* On $\{Y_{n+1} = c\}$, the counts formed in line 2 for candidate c equal the class-c stratum counts of $\mathcal{B}$, and the merge in line 3 depends only on these counts; hence Γ_c is a function of $\mathcal{B}$, and whichever class-c point of $\mathcal{B}$ occupies the test position, the same Γ_c results. *Conditioning.* Fix a bag value with $\Gamma_c = \gamma$ and a group $g \in \gamma$, and let M be the number of class-c points of $\mathcal{B}$ with stratum in g; the calibration count is $m = M - 1$. By exchangeability, conditional on $\mathcal{B}$ the test point is uniformly distributed over the elements of $\mathcal{B}$; conditional additionally on $\{Y_{n+1} = c,\ b(X_{n+1}) \in g\}$, it is uniformly distributed over these M points, each of which would be assigned to group g. *Rank argument.* Let R be the rank of the test score among the M scores, ties broken uniformly at random, so R is uniform on $\{1, \ldots, M\}$. With $k = \lceil M(1-\alpha) \rceil = \lceil (m+1)(1-\alpha) \rceil$: if $k \leq m$, the test point is covered whenever $R \leq k$, because its score then does not exceed the k-th smallest of the other m scores, so the conditional coverage is at least $k/M \geq 1 - \alpha$; if $k > m$, $\hat{q} = +\infty$ and coverage is one. *Integration.* On $\{Y_{n+1} = c\}$ the event $\{\Gamma_c = \gamma\}$ is a union of bag values, so integrating the bag-conditional bound over those bags gives the group statement; averaging over groups with weights $\mathbb{P}(\Gamma_c = \gamma, b(X) \in g \mid Y = c)$ gives $\mathrm{cov}_c \geq 1 - \alpha$. □

Remark 1 (Three limits of Theorem S1). (i) *Groups, not strata.* The conditioning event is the realized merged group. When strata merge, one constituent stratum can be under-covered while another compensates, exactly as Proposition S1 allows one level up; per-stratum validity belongs to fixed class × support calibration, and to merging only where it does not merge. All low-support results in the paper are reported per original stratum. (ii) *The budget is met by the bag.* A group closes when its bag count reaches n_{req}, so a single-stratum group can hold only $n_{\mathrm{req}} - 1$ calibration points; the tolerance statement below applies at $n \geq n_{\mathrm{req}}$. The constant n_{req} was frozen on 9 October 2026; only this wording is made precise. (iii) *Exchangeability.* The guarantee needs Assumption E and is claimed for random splits only.

A.5 Calibration budget and the design rule

For i.i.d. data with continuous scores, the coverage of a cell with n calibration points, conditional on those points, is a random quantity whose distribution over calibration draws is $\mathrm{Beta}(k_n, n + 1 - k_n)$ [9]. Before the confirmatory runs we fixed

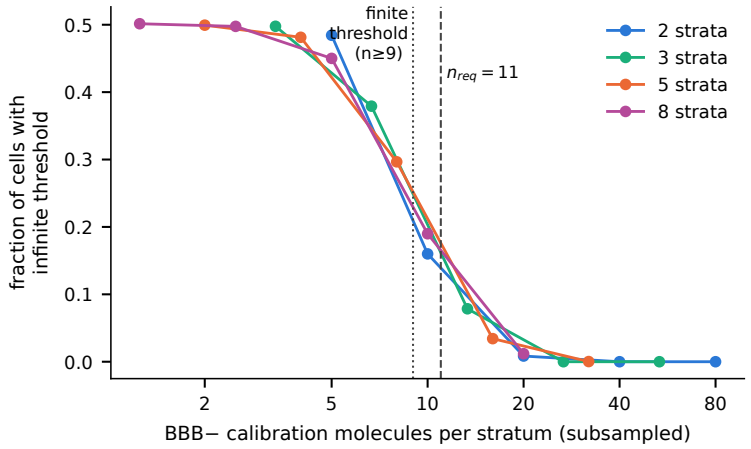
$$n_{\mathrm{req}}(\alpha, \epsilon, \delta) = \min\left\{n_0 :\ \forall n \geq n_0,\ \mathbb{P}\big(\mathrm{Beta}(k_n, n+1-k_n) < 1 - \alpha - \epsilon\big) \leq \delta\right\}, \qquad \epsilon = \alpha,\ \delta = 0.1.$$

The minimum over all $n \geq n_0$ is needed because k_n jumps and the tail probability is not monotone in n. Table S1 contrasts it with the finite-threshold condition $k_n \leq n$. It is a tolerance rule, not a certificate of

Table S1: Two calibration-size thresholds. Finite threshold: smallest n with $k_n \leq n$. n_{req}: tolerance rule above (with probability at least $1 - \delta = 0.9$ over calibration draws, realized cell coverage is at least $1 - 2\alpha$).

α	finite threshold	n_{req}
0.05	19	22
0.10	9	11
0.20	4	5

$1 - \alpha$ conditional coverage. Dividing the minority calibration set among B strata gives the planning heuristic $B \leq n_{\mathrm{cal}}^{-}/n_{\mathrm{req}}(\alpha)$. It assumes roughly equal strata and does not guarantee that every stratum of a given split reaches n_{req}; Fig. S1 shows that the fraction of cells with an infinite threshold is about one half when strata hold only a few minority calibration molecules and falls steeply as they approach and exceed n_{req}, for 2, 3, 5 and 8 strata alike, which is what the heuristic predicts.



BBB− calibration size	2 bins	3 bins	5 bins	8 bins
10	0.484 (5.0)	0.498 (3.3)	0.499 (2.0)	0.501 (1.2)
20	0.160 (10.0)	0.379 (6.7)	0.481 (4.0)	0.498 (2.5)
40	0.008 (20.0)	0.079 (13.3)	0.297 (8.0)	0.450 (5.0)
80	0.000 (40.0)	0.000 (26.7)	0.034 (16.0)	0.190 (10.0)
160	0.000 (80.0)	0.000 (53.3)	0.000 (32.0)	0.012 (20.0)
realized (65–221)	0.000	0.006	0.065	0.154

Figure S1: Feasibility of fixed class × support partitions against minority calibration molecules per stratum (BBB− calibration subsampled, 10 draws per split, majority retained; LAC, $\alpha = 0.1$, CE; mean over setting × backbone cells in which the size is reachable in all 20 splits). Dotted: finite-threshold condition; dashed: $n_{\text{req}} = 11$. Table: the plotted values; parentheses, BBB− calibration molecules per stratum; last row, the realized calibration sets without subsampling.

A.6 Exact invariances

Proposition S4 (Invariance to post-hoc per-class recalibration)**.** *If $\hat{p}'_y = h_y(\hat{p}_y)$ with each h_y strictly increasing, class-conditional and class × support LAC prediction sets are unchanged; marginal LAC sets are in general changed.*

Proof. Within class y, all calibration and test LAC scores are transformed by the same strictly increasing map $u \mapsto 1 - h_y(1-u)$, which preserves their order. Each per-class or per-cell threshold is an order statistic of class-y scores, and membership is the comparison $s(x,y) \le \hat{q}_y$, which the common map preserves. A marginal threshold is an order statistic of scores of both classes, transformed by different maps, so the ordering between classes, and hence the threshold, can change. □

Remark. For a binary model, temperature scaling applies the same increasing map to the logit of both classes, so $h_1(p) = 1 - h_0(1-p)$ and both classes' LAC scores share one map; marginal sets are then unchanged as well. Logit adjustment shifts the two classes differently. Empirically (per-split calibration results, 12 in-domain cells), temperature scaling and logit adjustment change class-conditional, class × support and merged sets by at most 0.0000 and 0.0002 in any endpoint, the nonzero value being a set-size difference caused by ties among probabilities stored in single precision near 1; marginal sets change by at most 0.0006 under temperature scaling and by up to 0.20 in BBB− coverage and 0.37 in low-support BBB− coverage under logit adjustment.

Proposition S5 (Margin and LAC coincide for binary tasks)**.** *For $K = 2$, the margin score $\max_{y' \neq y} \hat{p}_{y'} - \hat{p}_y = 1 - 2\hat{p}_y = 2s_{\text{LAC}}(x,y) - 1$, a strictly increasing affine function of LAC common to both labels; every calibrator whose threshold is an order statistic of scores therefore returns identical sets.*

A.7 Exact results versus design choices, and relation to prior theory

Table S2: Status of each formal statement.

Statement	Status	Scope
Stratum floor (Prop. S1)	exact	any class-conditionally valid predictor
Shared-threshold coverage (Prop. S2)	exact	I, C; ordering needs stochastic dominance
Cell coverage band (Prop. S3)	exact	fixed partition; E, C
Cell-size change (Cor. S1)	exact two-sided bound	class level; random strata
Merging validity (Thm. S1)	exact	realized merged groups; E
Beta law	exact [9]	i.i.d., continuous scores; over calibration draws
n_{req} and $B \leq n_{\text{cal}}^{-}/n_{\text{req}}$	design choice	tolerance $\epsilon = \alpha$, $\delta = 0.1$; heuristic
Invariances (Props. S4, S5)	exact	LAC; binary for the margin

No informative method can guarantee distribution-free coverage conditional on every subgroup [4], so practical methods condition on a finite taxonomy, and a subgroup outside it can be under-covered. That result does not identify the subgroup or the direction. Proposition S2 does so when scores are stochastically ordered across strata: the stratum with stochastically largest scores is the under-covered one. In these data a pre-assay covariate (support) identifies the under-covered subgroup, and Proposition S3 shows that joint conditioning on it removes the gradient at a cost bounded cell by cell. Clustered [2], Kandinsky [1], interpolated [3], localized [6, 7] and function-class-conditional [5] methods choose or relax the taxonomy differently; Section D compares them on the same predictions.

B Data, splits and models

Curation. SMILES were standardized (largest organic fragment, neutralization, canonical isomeric SMILES), de-duplicated by InChIKey with conflicting-label groups removed, and stereoisomer families sharing a connectivity key collapsed to one record when labels agreed and removed otherwise. This gives 1,924 BBBP molecules (23.5% BBB−) and 3,887 B3DB molecules (36.5% BBB−). The external transfer set contains the 2,119 B3DB molecules whose connectivity key does not occur in BBBP (1,070 BBB−).

Splits. TRAIN/VAL/CAL/TEST fractions are 60/10/15/15%. Random splits permute molecules uniformly. Balanced scaffold splits assign Bemis-Murcko scaffold groups at random, with large groups placed in TRAIN and acyclic molecules treated as singleton groups. Cluster holdout withholds whole k-means clusters of ECFP4 fingerprints ($k = 20$) as TEST. Transfer trains and calibrates on BBBP scaffold splits and tests on the external set. No split function receives labels.

Support. $s_{\max}$ is the maximal Tanimoto similarity of 2048-bit ECFP4 fingerprints (radius 2) to TRAIN, computed per split for VAL, CAL and TEST. Stratum edges are VAL quantiles. The CAL-edge variant is reported in Table S8.

Models. Random forest: 500 trees, $\sqrt{d}$ features, ECFP4 input. GIN: 5 layers, 128 hidden units, dropout 0.5, learning rate 10^{-3}, weight decay 10^{-5}, up to 60 epochs with patience 15 on VAL AUROC. Chemprop v2 D-MPNN: default bond message passing, mean aggregation, batch normalization, up to 40 epochs with patience 10 on VAL loss. wCE uses inverse class-frequency weights.

Predictive performance. Table S3 reports the held-out discrimination of each backbone, so that the coverage results can be read against model competence.

Pre-specification. The primary family (endpoint, operating point, settings, backbones, three contrasts, Holm over 36) was fixed in a dated amendment to the statistical analysis plan on 21 October 2026, before the confirmatory calibration outputs were analysed; n_{req} was frozen on 9 October 2026, before the confirmatory seeds were run.

Table S3: Predictive performance on TEST (mean and SD over 20 splits, CE). AUPRC and recall refer to the BBB− class; balanced accuracy and recall use the argmax decision.

Setting	Model	AUROC (SD)	AUPRC, BBB− (SD)	BBB− share	balanced acc.	BBB− recall
BBBP random	RF	0.911 (0.025)	0.817 (0.049)	0.24	0.795	0.615
BBBP random	GIN	0.897 (0.023)	0.780 (0.040)	0.24	0.798	0.657
BBBP random	Chemprop	0.910 (0.024)	0.791 (0.045)	0.24	0.813	0.688
BBBP scaffold	RF	0.882 (0.037)	0.773 (0.068)	0.25	0.734	0.490
BBBP scaffold	GIN	0.864 (0.057)	0.746 (0.079)	0.25	0.756	0.583
BBBP scaffold	Chemprop	0.869 (0.038)	0.728 (0.068)	0.25	0.741	0.545
B3DB random	RF	0.883 (0.011)	0.817 (0.020)	0.36	0.745	0.568
B3DB random	GIN	0.861 (0.018)	0.798 (0.027)	0.36	0.757	0.631
B3DB random	Chemprop	0.860 (0.015)	0.792 (0.026)	0.36	0.755	0.636
B3DB scaffold	RF	0.867 (0.021)	0.803 (0.029)	0.37	0.729	0.539
B3DB scaffold	GIN	0.846 (0.026)	0.790 (0.031)	0.37	0.743	0.615
B3DB scaffold	Chemprop	0.843 (0.037)	0.771 (0.050)	0.37	0.738	0.627
BBBP→B3DB	RF	0.717 (0.008)	0.728 (0.007)	0.50	0.579	0.190
BBBP→B3DB	GIN	0.647 (0.020)	0.678 (0.021)	0.50	0.598	0.348
BBBP→B3DB	Chemprop	0.686 (0.014)	0.688 (0.016)	0.50	0.597	0.299

C Additional results

Construction of main-text Figure 1. The figure is generated by `scripts/make_fig1_overview.py` as an editable diagrams.net file from the confirmatory caches, and every value it shows is recomputed by the independent verifier. *Setting.* For each of the twelve setting × backbone cells we took the mean high-minus-low BBB− coverage gap under class-conditional calibration (LAC, $\alpha = 0.1$, CE; 0.073–0.149) and selected the lower median among the six random-split cells, where exchangeability holds and the panels can be read against Section A: B3DB, random split, Chemprop, 20 splits, gap 0.116. The lower median of all twelve cells, BBBP scaffold split, GIN (gap 0.115), lies in BBBP scaffold, the one setting in which the primary repair is not significant, so it would illustrate a correction that the paper does not claim. The rule was fixed after the confirmatory results were known; the figure is an illustration, not additional evidence. *Panel B.* Each BBB− test appearance is shown as its score minus the class-conditional threshold of its own split, so that zero is the applicable threshold for every point; no single raw threshold is drawn across splits, and the vertical axis spans the full range 0 to 1. The 1461, 1359 and 1387 low-, mid- and high-support appearances come from 607, 599 and 555 distinct molecules; they are not independent, and the pooled curves are descriptive only (all inference in the paper is over splits). Of the pooled appearances, 15.4%, 8.7% and 3.7% lie above the threshold; one minus split-mean coverage in panel A gives 15.3%, 8.7% and 3.8%. The two differ by at most 0.1 percentage point, because pooling weights each split by its number of test appearances, and agree at the whole-percent precision printed in the panel. The panel restates the coverage arithmetic of Proposition S2 on these data; it does not identify why low-support compounds receive higher scores. Measured from the class threshold, the low-support class × support threshold lies between 0.014 and 0.120 and the high-support one between -0.369 and -0.043; the low stratum has 53–94 BBB− calibration compounds per split. Switching to class × support calibration covers 117 BBB− test appearances that class-conditional calibration excluded (99 low, 18 mid, 0 high support) and excludes 108 that it covered (0 low, 18 mid, 90 high), the redistribution of Propositions S2 and S3 seen in individual compounds. *Example pair.* Among BBB− test compounds in the low-support tertile at every appearance, we took the one with the most splits in which it is excluded under class-conditional and included under class × support calibration (ties: more appearances, then identifier). This is halofantrine: our identifier B3DB:2957, B3DB record 5199 ("halofantrine hydrochloride", PubChem CID 37393, BBB−, evidence group C, 1 reference), absent from BBBP. It is excluded and then included in 3 of its 5 test appearances, with maximal similarity to TRAIN 0.25–0.26 in all of them. In split 103, the first such split, its BBB− score is 0.908 against a class threshold of 0.904 (difference +0.004) and a low-support threshold of 0.918, set by 75 BBB− calibration compounds. Its most similar TRAIN compound in that split is B3DB record 650, 4-chlorobenzotrifluoride (our B3DB:538, CID 7394, BBB+), at Tanimoto 0.25 on RDKit Morgan fingerprints of radius 2 and 2048 bits, the fingerprint that defines support; the next most similar TRAIN compound is at 0.239. The pair is drawn in the design strip with native bonds and atom labels from RDKit 2D coordinates. Halofantrine is not marked in panel B, because a point +0.004 from the threshold would be visually indistinguishable from the threshold itself.

Table S4: Per-cell coverage with 95% intervals over 20 splits (LAC, $\alpha = 0.1$, CE). n^-_{cal}, n^-_{test}, n^-_{low}: mean BBB− calibration, test and low-support test counts per split. C×S: class × support.

Setting	Model	n^-_{cal}	n^-_{test}	n^-_{low}	Mondrian low	Mondrian high	C×S low	C×S high
BBBP random	RF	65	70	19	0.79 [0.74, 0.84]	0.94 [0.92, 0.96]	0.93 [0.89, 0.96]	0.89 [0.85, 0.93]
BBBP random	GIN	65	70	19	0.81 [0.74, 0.87]	0.95 [0.93, 0.97]	0.93 [0.90, 0.97]	0.91 [0.88, 0.94]
BBBP random	Chemprop	65	70	19	0.84 [0.78, 0.90]	0.97 [0.96, 0.98]	0.96 [0.93, 0.99]	0.91 [0.87, 0.95]
BBBP scaffold	RF	67	72	25	0.82 [0.76, 0.87]	0.96 [0.94, 0.98]	0.91 [0.87, 0.95]	0.92 [0.87, 0.96]
BBBP scaffold	GIN	67	72	25	0.84 [0.78, 0.90]	0.96 [0.92, 0.99]	0.92 [0.87, 0.97]	0.85 [0.77, 0.94]
BBBP scaffold	Chemprop	67	72	25	0.85 [0.79, 0.91]	0.97 [0.95, 0.99]	0.94 [0.90, 0.97]	0.88 [0.81, 0.95]
B3DB random	RF	210	210	73	0.87 [0.85, 0.89]	0.94 [0.93, 0.96]	0.93 [0.91, 0.94]	0.91 [0.89, 0.94]
B3DB random	GIN	210	210	73	0.86 [0.83, 0.88]	0.96 [0.94, 0.97]	0.92 [0.90, 0.94]	0.92 [0.90, 0.94]
B3DB random	Chemprop	210	210	73	0.85 [0.83, 0.87]	0.96 [0.95, 0.98]	0.91 [0.90, 0.93]	0.90 [0.88, 0.92]
B3DB scaffold	RF	221	216	79	0.84 [0.82, 0.87]	0.94 [0.92, 0.96]	0.90 [0.88, 0.92]	0.89 [0.86, 0.92]
B3DB scaffold	GIN	221	216	79	0.83 [0.80, 0.87]	0.94 [0.92, 0.96]	0.88 [0.84, 0.92]	0.89 [0.86, 0.92]
B3DB scaffold	Chemprop	221	216	79	0.85 [0.82, 0.87]	0.95 [0.93, 0.96]	0.89 [0.87, 0.92]	0.91 [0.88, 0.93]

Table S5: Pre-specified primary family (36 tests). Δ: mean paired difference in low-support BBB− coverage; seed CI: bootstrap over splits; p: Wilcoxon signed-rank; Holm_{36}: Holm-adjusted over the 36 tests. Molecule CI and Holm_{mol}, $\text{Holm}_{\text{scaf}}$: bootstrap over unique test molecules or scaffolds (20000 draws, calibration fixed), Holm over 36.

Setting	Model	Contrast	Δ	Seed CI	p	Holm_{36}	Molecule CI	Holm_{mol}	$\text{Holm}_{\text{scaf}}$
B3DB random	Chemprop	C×S - Mondrian	+0.066	[0.050, 0.084]	<0.001	0.004	[0.052, 0.081]	0.004	0.004
B3DB random	Chemprop	SABM - Mondrian	+0.066	[0.050, 0.084]	<0.001	0.004	[0.052, 0.081]	0.004	0.004
B3DB random	Chemprop	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
B3DB random	GIN	C×S - Mondrian	+0.063	[0.050, 0.077]	<0.001	<0.001	[0.050, 0.078]	0.004	0.004
B3DB random	GIN	SABM - Mondrian	+0.063	[0.050, 0.077]	<0.001	<0.001	[0.050, 0.078]	0.004	0.004
B3DB random	GIN	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
B3DB random	RF	C×S - Mondrian	+0.059	[0.039, 0.080]	<0.001	0.004	[0.045, 0.074]	0.004	0.004
B3DB random	RF	SABM - Mondrian	+0.059	[0.039, 0.080]	<0.001	0.004	[0.045, 0.074]	0.004	0.004
B3DB random	RF	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
B3DB scaffold	Chemprop	C×S - Mondrian	+0.048	[0.032, 0.066]	<0.001	0.004	[0.037, 0.059]	0.004	0.004
B3DB scaffold	Chemprop	SABM - Mondrian	+0.048	[0.032, 0.066]	<0.001	0.004	[0.037, 0.059]	0.004	0.004
B3DB scaffold	Chemprop	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
B3DB scaffold	GIN	C×S - Mondrian	+0.045	[0.028, 0.062]	<0.001	0.012	[0.035, 0.057]	0.004	0.004
B3DB scaffold	GIN	SABM - Mondrian	+0.045	[0.028, 0.062]	<0.001	0.012	[0.035, 0.057]	0.004	0.004
B3DB scaffold	GIN	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
B3DB scaffold	RF	C×S - Mondrian	+0.054	[0.036, 0.076]	<0.001	0.005	[0.042, 0.067]	0.004	0.004
B3DB scaffold	RF	SABM - Mondrian	+0.054	[0.036, 0.076]	<0.001	0.005	[0.042, 0.067]	0.004	0.004
B3DB scaffold	RF	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
BBBP random	Chemprop	C×S - Mondrian	+0.120	[0.074, 0.168]	<0.001	0.006	[0.082, 0.161]	0.004	0.004
BBBP random	Chemprop	SABM - Mondrian	+0.120	[0.074, 0.168]	<0.001	0.006	[0.082, 0.161]	0.004	0.004
BBBP random	Chemprop	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
BBBP random	GIN	C×S - Mondrian	+0.126	[0.082, 0.173]	<0.001	0.005	[0.087, 0.168]	0.004	0.004
BBBP random	GIN	SABM - Mondrian	+0.126	[0.082, 0.173]	<0.001	0.005	[0.087, 0.168]	0.004	0.004
BBBP random	GIN	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
BBBP random	RF	C×S - Mondrian	+0.137	[0.089, 0.187]	<0.001	0.005	[0.096, 0.181]	0.004	0.004
BBBP random	RF	SABM - Mondrian	+0.137	[0.089, 0.187]	<0.001	0.005	[0.096, 0.181]	0.004	0.004
BBBP random	RF	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
BBBP scaffold	Chemprop	C×S - Mondrian	+0.083	[0.032, 0.141]	0.005	0.077	[0.053, 0.116]	0.004	0.004
BBBP scaffold	Chemprop	SABM - Mondrian	+0.083	[0.032, 0.141]	0.005	0.077	[0.053, 0.116]	0.004	0.004
BBBP scaffold	Chemprop	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
BBBP scaffold	GIN	C×S - Mondrian	+0.082	[0.032, 0.137]	0.004	0.077	[0.053, 0.113]	0.004	0.004
BBBP scaffold	GIN	SABM - Mondrian	+0.082	[0.032, 0.137]	0.004	0.077	[0.053, 0.113]	0.004	0.004
BBBP scaffold	GIN	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000
BBBP scaffold	RF	C×S - Mondrian	+0.091	[0.042, 0.143]	0.004	0.077	[0.063, 0.121]	0.004	0.004
BBBP scaffold	RF	SABM - Mondrian	+0.091	[0.042, 0.143]	0.004	0.077	[0.063, 0.121]	0.004	0.004
BBBP scaffold	RF	SABM - C×S	+0.000	[0.000, 0.000]	1.000	1.000	[0.000, 0.000]	1.000	1.000

Table S6: Empirical settings (LAC, $\alpha = 0.1$, CE; mean over 20 splits). No finite-sample guarantee applies.

Setting	Model	Marg. BBB−	Mond. BBB−	Marg. BBB+	Mond. BBB+	Mond. low	C×S low	Size M	Size C×S
BBBP→B3DB	RF	0.38	0.86	0.98	0.77	0.84	0.90	1.36	1.45
BBBP→B3DB	GIN	0.50	0.77	0.92	0.81	0.74	0.88	1.41	1.51
BBBP→B3DB	Chemprop	0.51	0.81	0.95	0.79	0.78	0.88	1.36	1.46
BBBP cluster	RF	0.54	0.92	0.99	0.79	0.94	0.98	1.33	1.46
BBBP cluster	GIN	0.69	0.82	0.93	0.83	0.84	0.93	1.25	1.38
BBBP cluster	Chemprop	0.65	0.84	0.93	0.84	0.85	0.89	1.26	1.36
B3DB cluster	RF	0.74	0.92	0.98	0.79	0.92	0.95	1.45	1.52
B3DB cluster	GIN	0.74	0.83	0.93	0.87	0.81	0.86	1.38	1.48
B3DB cluster	Chemprop	0.74	0.84	0.92	0.85	0.83	0.89	1.36	1.48

Table S7: Score and level sensitivity. Low-support BBB− coverage, mean (range) over the 12 in-domain cells (six random-split and six scaffold-split), CE.

Score, level	Mondrian low	C×S low	cells with gain > 0
LAC, $\alpha = 0.10$ (primary)	0.837 (0.79–0.87)	0.918 (0.88–0.96)	12/12
APS, $\alpha = 0.10$	0.847 (0.82–0.88)	0.915 (0.89–0.95)	12/12
RAPS, $\alpha = 0.10$	0.852 (0.81–0.87)	0.918 (0.88–0.95)	12/12
LAC, $\alpha = 0.05$	0.928 (0.91–0.95)	0.967 (0.94–0.99)	12/12
LAC, $\alpha = 0.20$	0.687 (0.59–0.75)	0.816 (0.76–0.88)	12/12

Table S8: Second Mondrian dimension. Low-support BBB− coverage evaluated on the common VAL tertiles and mean set size, over the 12 in-domain cells (six random-split and six scaffold-split) (LAC, $\alpha = 0.1$, CE).

Second Mondrian dimension	low-support BBB− (mean)	range	set size
class only (Mondrian)	0.837	0.79–0.87	1.285
max-Tanimoto tertiles (primary)	0.918	0.88–0.96	1.329
max-Tanimoto, 2 bins	0.908	0.87–0.95	1.308
max-Tanimoto, 5 bins	0.928	0.88–0.97	1.381
mean top-5 Tanimoto tertiles	0.905	0.86–0.93	1.326
MACCS max-Tanimoto tertiles	0.892	0.85–0.92	1.333
k-means clusters ($k = 3$)	0.855	0.83–0.88	1.390
TPSA strata (< 60, 60–90, > 90)	0.868	0.84–0.89	1.501
tertile edges from CAL	0.913	0.86–0.94	1.326
random strata (control)	0.860	0.83–0.89	1.335

Table S9: Same-cell-size control (exploratory). Low- and high-support BBB− coverage under class-conditional calibration, class × randomly assigned strata of equal size, and class × Tanimoto tertiles (C×S), with mean set sizes; paired difference in low-support coverage, seed bootstrap interval, and Holm-adjusted Wilcoxon p over the 12 cells. LAC, $\alpha = 0.1$, CE, 20 splits.

Setting	Model	Mondrian low/high	random strata low/high	C×S low/high	size rand./C×S	C×S − random, low	Holm p
B3DB random	Chemprop	0.85 / 0.96	0.85 / 0.97	0.91 / 0.90	1.35 / 1.33	+0.059 [0.043, 0.075]	0.001
B3DB random	GIN	0.86 / 0.96	0.86 / 0.96	0.92 / 0.92	1.35 / 1.35	+0.055 [0.042, 0.069]	0.001
B3DB random	RF	0.87 / 0.94	0.89 / 0.95	0.93 / 0.91	1.28 / 1.29	+0.043 [0.026, 0.060]	0.005
B3DB scaffold	Chemprop	0.85 / 0.95	0.86 / 0.95	0.89 / 0.91	1.35 / 1.35	+0.036 [0.017, 0.054]	0.014
B3DB scaffold	GIN	0.83 / 0.94	0.84 / 0.94	0.88 / 0.89	1.35 / 1.35	+0.037 [0.017, 0.056]	0.013
B3DB scaffold	RF	0.84 / 0.94	0.86 / 0.94	0.90 / 0.89	1.30 / 1.30	+0.040 [0.022, 0.059]	0.008
BBBP random	Chemprop	0.84 / 0.97	0.88 / 0.98	0.96 / 0.91	1.32 / 1.29	+0.084 [0.048, 0.123]	0.004
BBBP random	GIN	0.81 / 0.95	0.84 / 0.95	0.93 / 0.91	1.29 / 1.30	+0.090 [0.050, 0.133]	0.008
BBBP random	RF	0.79 / 0.94	0.83 / 0.96	0.93 / 0.89	1.27 / 1.29	+0.100 [0.061, 0.140]	0.005
BBBP scaffold	Chemprop	0.85 / 0.97	0.86 / 0.97	0.94 / 0.88	1.37 / 1.36	+0.073 [0.029, 0.124]	0.020
BBBP scaffold	GIN	0.84 / 0.96	0.89 / 0.98	0.92 / 0.85	1.41 / 1.38	+0.037 [-0.003, 0.075]	0.059
BBBP scaffold	RF	0.82 / 0.96	0.86 / 0.97	0.91 / 0.92	1.37 / 1.35	+0.046 [-0.003, 0.097]	0.099

D Mechanism and existing conditional methods (exploratory)

Existing methods. Table S12 evaluates published conditional, localized and reweighted conformal methods on the cached predictions of the 12 in-domain cells (six random-split and six scaffold-split). Several are our own implementations or approximations of the published ideas, as stated in the table; they are included to test whether the support gradient survives methods designed for conditional coverage, not to rank those methods.

Mechanism. For every split, the mean predicted probability of the true class was computed by support tertile, separately for BBB− and BBB+ test compounds (Table S13). Three predictions were stated before the confirmatory seeds were examined: (P1) low-support BBB+ compounds are covered at target while low-support BBB− compounds are not; (P2) the true-class probability falls with support more for BBB− than for BBB+; (P3) in the low-support tertile, the mean predicted BBB− probability is below the observed BBB− fraction. P2 holds in 12 of 12 cells. P1 fails: low-support BBB+ coverage is 0.85–0.89, and the lower end of its 95% interval is at or above 0.88 (a tolerance of 0.02 below target, fixed in advance) in 0 of 12 cells. P3 holds in 6 of 12 cells (4 of 6 on B3DB).

Continuous support. Over all BBB− test appearances, the logistic slope of class-conditional coverage on maximal Tanimoto support is 0.21–0.48 log-odds per 0.1, with a split-bootstrap interval above zero in 12 of 12 cells; under class × support it is -0.16 to 0.13 and its interval includes zero in most cells (Fig. S4).

Calibration at a fixed score. A logistic regression of the BBB− label on the logit of the predicted

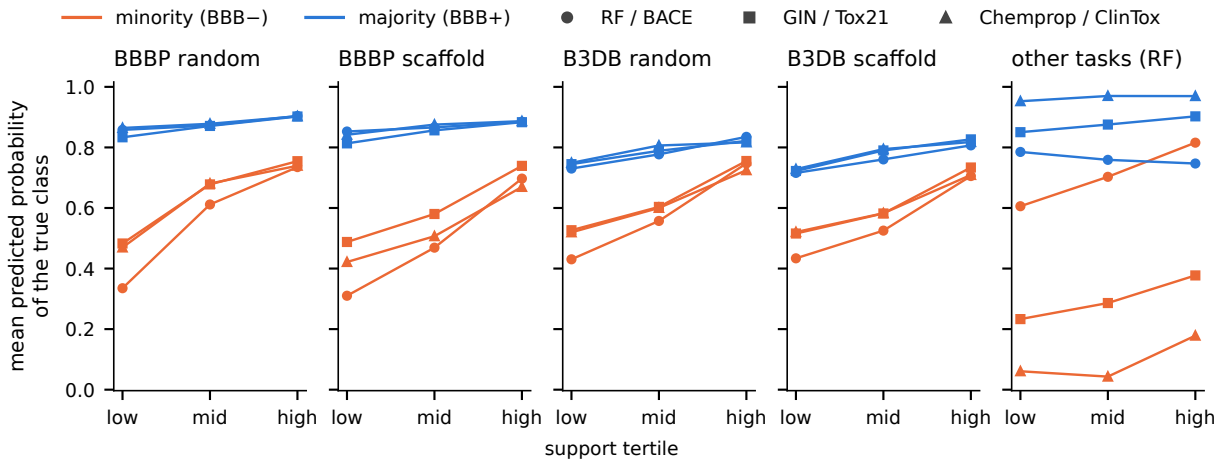


Figure S2: Mean predicted probability of the true class by support tertile for minority (orange) and majority (blue) test compounds under each backbone (markers), LAC, CE, 20 splits. Right: random forest on the additional tasks (circles BACE, squares Tox21 SR-ARE, triangles ClinTox, random splits). The minority curve falls more steeply than the majority curve in every BBB cell; on Tox21 SR-ARE it also falls, yet the coverage gap is reversed (Proposition S2(c)).

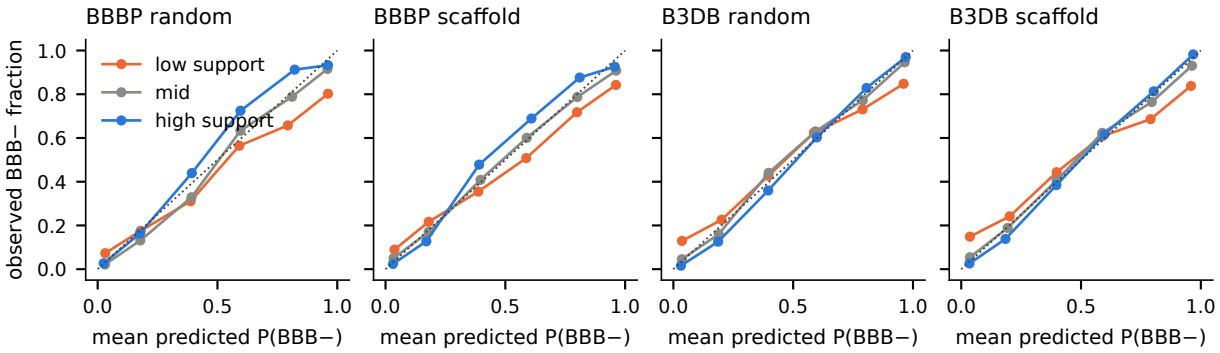


Figure S3: Reliability by support tertile. Observed BBB− fraction against mean predicted BBB− probability in fixed probability bins (at least 30 test appearances per point), pooled over backbones and 20 splits. On B3DB, low-support compounds (orange) are BBB− more often than predicted at low predicted probability; on BBBP the pattern is inconsistent, as the support coefficient indicates.

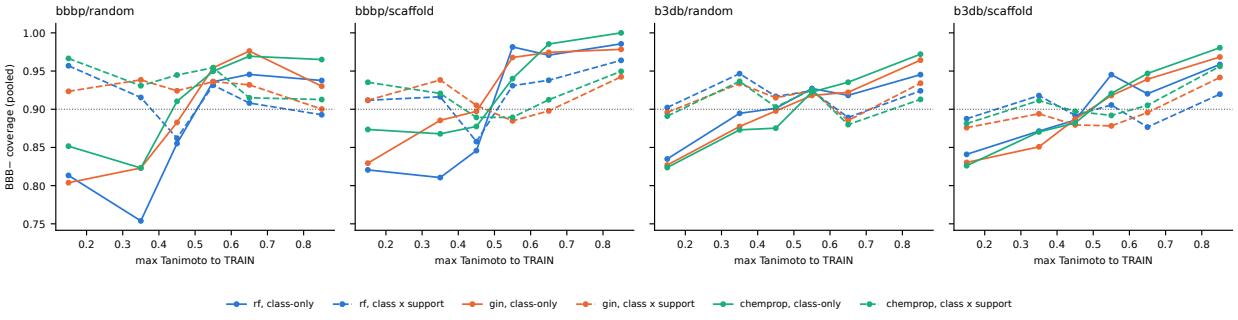


Figure S4: BBB− coverage as a function of maximal Tanimoto support (LAC, $\alpha = 0.1$, CE), pooled over 20 splits in fixed support bins with at least 20 appearances. Solid: class-conditional; dashed: class × support. Pooled intervals are optimistic because molecules recur across splits; inference uses the split bootstrap of the slope.

Table S10: Common evaluation grid. Every partition scored on the lowest VAL tertile (low-tertile cov.) and on the worst VAL sextile with at least five BBB− molecules; BBB− calibration subsampled (10 draws per split), majority retained; LAC, $\alpha = 0.1$, CE. Cells: setting × backbone combinations reachable in all 20 splits. Degenerate: fraction of (class, stratum) cells with an infinite threshold.

n_{cal}^{-}	Partition	cells	low-tertile cov.	worst sextile	set size	singleton	degenerate
realized	mondrian	12	0.837	0.764	1.285	0.715	0.000
realized	fixed-b2	12	0.908	0.783	1.308	0.688	0.000
realized	fixed-b3	12	0.918	0.787	1.329	0.662	0.006
realized	fixed-b5	12	0.928	0.817	1.381	0.611	0.065
realized	fixed-b8	12	0.940	0.833	1.471	0.520	0.154
realized	sabm-b3	12	0.918	0.788	1.324	0.668	0.000
realized	sabm-b8	12	0.923	0.796	1.374	0.616	0.000
160	mondrian	6	0.849	0.798	1.310	0.690	0.000
160	fixed-b2	6	0.896	0.809	1.325	0.675	0.000
160	fixed-b3	6	0.905	0.809	1.334	0.664	0.000
160	fixed-b5	6	0.912	0.824	1.359	0.638	0.000
160	fixed-b8	6	0.921	0.832	1.401	0.596	0.012
160	sabm-b3	6	0.905	0.809	1.334	0.664	0.000
160	sabm-b8	6	0.920	0.830	1.392	0.604	0.000
80	mondrian	6	0.851	0.801	1.315	0.685	0.000
80	fixed-b2	6	0.902	0.809	1.340	0.660	0.000
80	fixed-b3	6	0.914	0.812	1.361	0.636	0.000
80	fixed-b5	6	0.925	0.833	1.419	0.577	0.034
80	fixed-b8	6	0.940	0.848	1.523	0.473	0.190
80	sabm-b3	6	0.914	0.813	1.361	0.636	0.000
80	sabm-b8	6	0.924	0.819	1.402	0.594	0.000
40	mondrian	12	0.832	0.760	1.289	0.711	0.000
40	fixed-b2	12	0.924	0.795	1.354	0.640	0.008
40	fixed-b3	12	0.935	0.807	1.419	0.573	0.079
40	fixed-b5	12	0.959	0.860	1.568	0.424	0.297
40	fixed-b8	12	0.987	0.945	1.703	0.293	0.450
40	sabm-b3	12	0.921	0.792	1.357	0.637	0.000
40	sabm-b8	12	0.922	0.803	1.396	0.597	0.000
20	mondrian	12	0.839	0.771	1.311	0.688	0.000
20	fixed-b2	12	0.944	0.829	1.466	0.526	0.160
20	fixed-b3	12	0.975	0.897	1.628	0.366	0.379
20	fixed-b5	12	0.996	0.984	1.724	0.274	0.481
20	fixed-b8	12	0.999	0.995	1.754	0.245	0.498
20	sabm-b3	12	0.839	0.771	1.317	0.681	0.000
20	sabm-b8	12	0.839	0.771	1.336	0.662	0.000
10	mondrian	12	0.845	0.783	1.338	0.659	0.000
10	fixed-b2	12	0.997	0.989	1.720	0.280	0.484
10	fixed-b3	12	1.000	0.997	1.737	0.263	0.498
10	fixed-b5	12	1.000	1.000	1.745	0.255	0.499
10	fixed-b8	12	1.000	1.000	1.760	0.240	0.501
10	sabm-b3	12	0.845	0.783	1.344	0.652	0.000
10	sabm-b8	12	0.845	0.783	1.363	0.633	0.000

BBB− probability and support gives a support coefficient of -0.14 to -0.05 per 0.1 Tanimoto on B3DB, with intervals below zero in 6 of 6 cells, and -0.20 to 0.09 on BBBP, where signs differ across backbones.

Can calibration data alone flag the failure? Leave-one-out coverage of BBB− calibration molecules in the lowest support tertile, computed without test data, correlates weakly with test low-support coverage across the 240 splits (Spearman 0.11); as a per-split flag at 0.85 it has sensitivity 0.56 and specificity 0.47. Aggregated over splits, the calibration-only flag agrees with the test outcome in 10 of 12 cells, and the sign of the high-minus-low gap agrees in a fraction 0.85 of splits. Calibration data can therefore support an audit

Table S11: Training loss. Marginal BBB− coverage, class-conditional low-support coverage and class-conditional set size under CE and wCE (LAC, $\alpha = 0.1$).

Setting	Model	Marg. CE	Marg. wCE	Mond. low CE	Mond. low wCE	Size CE	Size wCE
BBBP random	RF	0.65	0.65	0.79	0.83	1.17	1.16
BBBP random	GIN	0.72	0.89	0.81	0.81	1.23	1.26
BBBP random	Chemprop	0.75	0.80	0.84	0.84	1.23	1.21
BBBP scaffold	RF	0.62	0.62	0.82	0.82	1.28	1.27
BBBP scaffold	GIN	0.69	0.87	0.84	0.86	1.34	1.35
BBBP scaffold	Chemprop	0.71	0.75	0.85	0.85	1.32	1.32
B3DB random	RF	0.79	0.79	0.87	0.87	1.26	1.27
B3DB random	GIN	0.82	0.90	0.86	0.85	1.33	1.32
B3DB random	Chemprop	0.81	0.85	0.85	0.87	1.32	1.33
B3DB scaffold	RF	0.77	0.77	0.84	0.84	1.29	1.29
B3DB scaffold	GIN	0.81	0.86	0.83	0.83	1.33	1.32
B3DB scaffold	Chemprop	0.82	0.85	0.85	0.86	1.33	1.36

Table S12: Existing conditional conformal methods on the same predictions (LAC, $\alpha = 0.1$, CE; 12 in-domain cells (six random-split and six scaffold-split), 20 splits each). Low-support BBB−: range over cells of the seed-mean coverage in the lowest VAL Tanimoto tertile; high: mean over cells for the highest tertile; cells failing: cells whose 95% interval for low-support coverage lies below 0.90; set size: mean over cells.

Method	Implementation	low-support BBB−	high	cells failing	set size
marginal split CP	standard	0.30–0.76	0.87	12	1.19
class-conditional (Mondrian)	standard	0.79–0.87	0.95	10	1.29
class x support, fixed tertiles	standard Mondrian taxonomy	0.88–0.96	0.90	0	1.33
Mondrian, training-conditional (PAC)	Vovk 2012 rank rule	0.86–0.92	0.97	1	1.40
Mondrian, SSBC-style Beta rank	our implementation	0.86–0.92	0.97	1	1.40
clustered CP over class x support cells	our implementation of Ding et al. 2023	0.68–0.89	0.84	8	1.25
Kandinsky-style group QR	our approximation, not the authors' algorithm	0.89–0.95	0.92	0	1.47
conditional QR on support	our approximation of Gibbs et al. 2025	0.88–0.94	0.91	0	1.33
INTERP-Q (q=0.5)	our implementation of Ding et al. 2026	0.52–0.81	0.91	12	1.22
INTERP-Q (q=0.9)	our implementation of Ding et al. 2026	0.73–0.85	0.95	12	1.26
RC3P	our implementation	0.79–0.87	0.95	10	1.31
localized CP, Tanimoto kernel	our implementation of Guan 2023	0.86–0.98	0.97	1	1.47
localized CP, kNN	our implementation	0.80–0.89	0.95	8	1.33
randomly localized CP	our implementation of Hore and Barber 2025	0.86–0.94	0.92	1	1.32
weighted Mondrian (density ratio)	our approximation	0.78–0.91	0.96	7	1.34
KMM-weighted Mondrian	our approximation, not released KMM-CP	0.76–0.87	0.95	11	1.28
macro-coverage CP	our implementation	0.62–0.80	0.92	12	1.31

across repeated splits but not a decision for a single deployment (Fig. S5).

Table S13: True-class probability and coverage by support under class-conditional calibration (LAC, $\alpha = 0.1$, CE; 20 splits). $\bar{p}_{\text{true}}$: mean predicted probability of the true class in the highest and lowest support tertile; cov.: low-support coverage of minority and majority compounds; calib. gap: mean predicted minority probability minus observed minority fraction in the lowest tertile, with 95% interval. Lower block: random forest on the additional tasks (Section E).

Setting	Model	$\bar{p}_{\text{true}}$ minority, high→low	$\bar{p}_{\text{true}}$ majority, high→low	cov. min. low	cov. maj. low	calib. gap low
BBBP random	RF	0.74 → 0.34	0.90 → 0.86	0.79	0.87	-0.02 [-0.04, -0.00]
BBBP random	GIN	0.75 → 0.48	0.90 → 0.83	0.81	0.86	0.03 [0.01, 0.05]
BBBP random	Chemprop	0.74 → 0.47	0.90 → 0.86	0.84	0.88	0.00 [-0.02, 0.02]
BBBP scaffold	RF	0.70 → 0.31	0.88 → 0.85	0.82	0.87	-0.06 [-0.09, -0.02]
BBBP scaffold	GIN	0.74 → 0.49	0.88 → 0.81	0.84	0.86	0.02 [-0.02, 0.05]
BBBP scaffold	Chemprop	0.67 → 0.42	0.89 → 0.84	0.85	0.85	-0.02 [-0.05, 0.01]
B3DB random	RF	0.75 → 0.43	0.83 → 0.73	0.87	0.89	-0.05 [-0.06, -0.03]
B3DB random	GIN	0.76 → 0.53	0.82 → 0.74	0.86	0.89	-0.02 [-0.04, -0.00]
B3DB random	Chemprop	0.72 → 0.52	0.82 → 0.75	0.85	0.87	-0.03 [-0.05, 0.00]
B3DB scaffold	RF	0.71 → 0.43	0.81 → 0.72	0.84	0.88	-0.05 [-0.07, -0.03]
B3DB scaffold	GIN	0.73 → 0.52	0.83 → 0.72	0.83	0.87	-0.03 [-0.06, 0.01]
B3DB scaffold	Chemprop	0.71 → 0.52	0.82 → 0.73	0.85	0.85	-0.03 [-0.05, -0.00]
BACE random	RF	0.82 → 0.61	0.75 → 0.79	0.85	0.95	-0.00 [-0.02, 0.02]
BACE scaffold	RF	0.73 → 0.55	0.77 → 0.73	0.82	0.92	-0.02 [-0.05, 0.02]
ClinTox random	RF	0.18 → 0.06	0.97 → 0.95	0.98	0.85	-0.03 [-0.05, -0.02]
ClinTox scaffold	RF	0.12 → 0.06	0.97 → 0.95	0.98	0.86	-0.03 [-0.04, -0.02]
Tox21 SR-ARE random	RF	0.38 → 0.23	0.90 → 0.85	0.92	0.89	-0.03 [-0.03, -0.02]
Tox21 SR-ARE scaffold	RF	0.34 → 0.24	0.90 → 0.83	0.93	0.89	-0.01 [-0.02, -0.00]

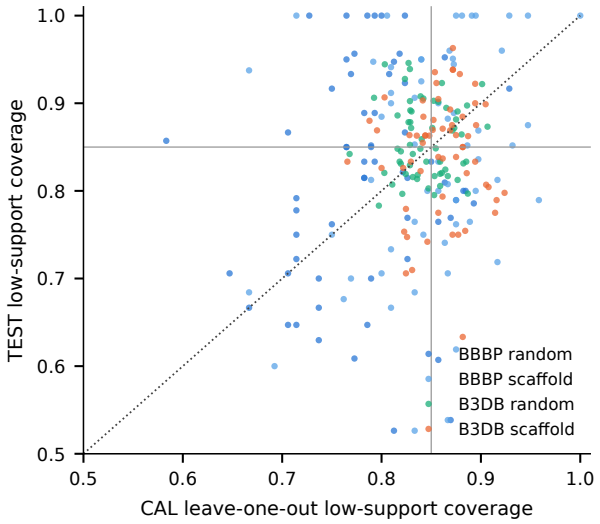


Figure S5: Calibration-only audit. Leave-one-out coverage of low-support BBB− calibration molecules against test low-support coverage, one point per split and backbone (240 points). Grey lines: flag threshold 0.85.

E Other tasks and the prior protocol (exploratory)

The same pipeline (random forest, CE, VAL tertile edges, seeds 100–119) was run on BACE, ClinTox and Tox21 SR-ARE from MoleculeNet [10], with the minority label read from each dataset, and the random forest and graph convolutional network of Tursunbadalov and Tursunbadalov [8] were rerun on their BBBP protocol (all parseable records, random 50/25/25 split, no VAL, so tertile edges come from CAL). Holm correction is applied within each block.

Table S14: Other imbalanced tasks and the prior BBBP protocol. Minority coverage (LAC, $\alpha = 0.1$, CE; mean over 20 splits); $n_{\text{cal}}^{\text{min}}$: minority calibration molecules per split; last two columns: paired gain of class × support over class-conditional calibration in low-support minority coverage, seed bootstrap interval and Holm p within the block. ClinTox cells are dominated by infinite thresholds and are not interpretable.

Task	Model	$n_{\text{cal}}^{\text{min}}$	Marg. class	Mond. class	Mond. low	Mond. high	C×S low	C×S − Mond., low	Holm p
BACE random	RF	99.3	0.91	0.91	0.85	0.94	0.93	+0.081 [0.045, 0.120]	0.004
BACE scaffold	RF	94.7	0.88	0.88	0.82	0.90	0.90	+0.081 [0.044, 0.122]	0.016
ClinTox random	RF	8.5	0.00	0.97	0.98	0.94	1.00	+0.023 [0.000, 0.057]	0.849
ClinTox scaffold	RF	8.3	0.00	0.99	0.98	1.00	1.00	+0.020 [0.000, 0.050]	0.849
Tox21 SR-ARE random	RF	132.0	0.36	0.90	0.92	0.87	0.90	-0.013 [-0.031, 0.003]	0.686
Tox21 SR-ARE scaffold	RF	148.3	0.46	0.91	0.93	0.87	0.91	-0.017 [-0.036, 0.000]	0.679
BBBP random 50/25/25	RF	119.2	0.61	0.90	0.82	0.93	0.91	+0.091 [0.048, 0.135]	<0.001
BBBP random 50/25/25	GCN	119.2	0.64	0.91	0.81	0.95	0.91	+0.095 [0.071, 0.120]	<0.001

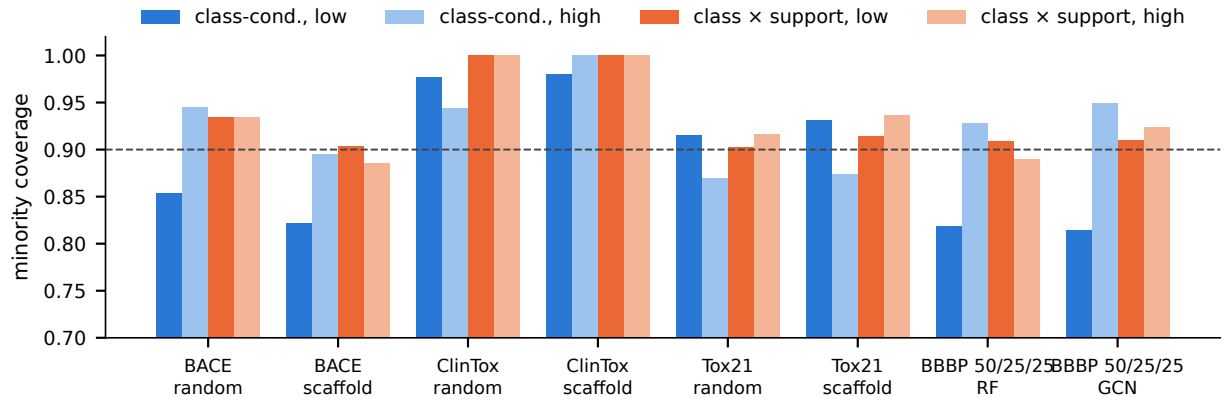


Figure S6: Low- and high-support minority coverage on other tasks and under the prior BBBP protocol (LAC, $\alpha = 0.1$, CE, 20 splits). BACE and the prior protocol show the BBB pattern (low below high under class-conditional calibration, levelled by class × support); Tox21 SR-ARE shows the reverse; ClinTox is dominated by infinite thresholds.

F Persistent failures

Definitions are given in the main text (Section 5). Table S16 lists every cross-architecture persistent failure under the strict definition, Fig. S7 shows their structures, and Table S17 gives descriptor contrasts under both definitions.

Curation findings. Record numbers refer to the `NO.` column of the B3DB classification file; identifiers in brackets are those used in our code. (i) Record 3838 [B3DB:2544], labelled BBB− (references R16, R10), is named rifampin but its SMILES encodes an unrelated difluoropyridine amide (PubChem CID 105381226); rifampin itself appears in records 1981, 5510 and 7390. (ii) Record 5692 [B3DB:3330], labelled BBB− from a single reference (R23), is named phenylbutazone but drawn with a propyl chain; record 279, the correctly drawn butyl compound, is labelled BBB+ (logBB −0.5, 25 references). (iii) Record 5253 [B3DB:3001], cyclobenzaprine hydrochloride, labelled BBB− from a single reference (R14), is drawn in a non-aromatic form that does not de-duplicate against record 1009, cyclobenzaprine, labelled BBB+ (logBB 1.06, three references); cyclobenzaprine is a centrally acting muscle relaxant. These are structural or record-level errors that can be checked from the file; they are distinct from disagreements between datasets about a correctly drawn compound, for which we do not decide which label is right.

Name-level screen. 33–38 eligible BBB− molecules per B3DB setting share a normalized compound name with a BBB+ record of the raw file. Among strict failures this holds for 1 (random) and 2 (scaffold), 10% and 14% against 3% and 3% of other eligible molecules (Fisher $p = 0.25$ and 0.07); under the majority rule 5% and 5% ($p = 0.34$, 0.25). Names are an imperfect key in both directions: in the raw B3DB file some compound names are attached to unrelated structures (for example, records named dihydroergotamine, polythiazide and demeclocycline), so a same-name match can join different compounds. This is therefore a

screen rather than a verdict on any label. The names given for shared failures in Table S16 were checked against their structures; only the rifampin record is mismatched, and it is flagged.

Label audit. Of 1,768 BBBP molecules found in B3DB by connectivity key (91.9%), 30 carry different labels (21 BBB− in BBBP and BBB+ in B3DB; 9 the reverse). Among eligible BBBP test molecules that occur in B3DB, the majority-rule failure rate is higher for label conflicts on random splits (3 of 18; odds ratio 6.7, Fisher $p = 0.024$) and scaffold splits (3 of 21; 8.0, $p = 0.017$). None of the 7 conflicting B3DB molecules is a failure.

Sensitivity without the three records. Removing the three records from calibration and test and recomputing all thresholds gives 9 cross-architecture failures on B3DB random (null mean 0.09) and 12 on B3DB scaffold (null mean 0.20), compared with 10 and 14 in the primary analysis; BBBP results are unaffected. The concordance conclusion is unchanged.

Concordance and appearance counts. Table S15 compares the observed number of strict cross-architecture failures with the primary null (each backbone's persistence indicator permuted within strata of the number of appearances k) and with a covariate-stratified null that also permutes within support and TPSA tertiles, which absorbs agreement explained by those covariates. It also gives the distribution of k: a strict failure tested twice is weaker evidence than one tested five times.

Table S15: Cross-architecture concordance against two nulls (10,000 permutations each) and appearance counts of strict failures. null_k: strata of k (capped at 6); null_{cov}: strata of $k\times$ support tertile $\times$ TPSA tertile.

Setting	eligible	strict	null_k	p_k	null_{cov}	p_{cov}	k: median (range)	$k \geq 3$	$k \geq 4$
B3DB random	1149	10	0.09	<0.001	0.77	<0.001	3 (2–5)	8	1
B3DB scaffold	1199	14	0.23	<0.001	0.83	<0.001	2 (2–5)	6	3
BBBP random	368	2	0.02	<0.001	0.32	0.018	4 (3–6)	2	1
BBBP scaffold	386	1	0.02	0.020	0.12	0.120	2 (2–2)	0	0
BBBP→B3DB	1070	0	0.01	1.000	0.03	1.000	–	0	0

Table S16: Cross-architecture persistent BBB− failures (strict definition). k: number of splits in which the molecule was tested; support: mean maximal Tanimoto similarity to TRAIN; flag: label in the other dataset or curation finding.

Setting	Compound	k	TPSA	HBD	MW	cLogP	support	flag
B3DB random	naftifine	5	3.2	0	287	4.99	0.53	
B3DB random	cyclobenzaprine hydrochloride	3	3.2	0	275	2.64	0.54	curation error (mis-drawn)
B3DB random	desloratadine	2	24.9	1	311	4.02	0.43	
B3DB random	bromhexine	3	29.3	1	376	4.56	0.29	
B3DB random	cytisine	3	34.0	1	190	0.55	0.61	
B3DB random	menadione	3	34.1	0	172	2.01	0.29	
B3DB random	dapiprazole	3	37.2	0	325	2.29	0.40	
B3DB random	B3DB:2805	3	39.1	1	255	2.53	0.29	
B3DB random	emetine hydrochloride	3	52.2	1	481	4.94	0.45	
B3DB random	troxipide	2	68.8	2	294	1.19	0.39	
B3DB scaffold	cyclobenzaprine hydrochloride	4	3.2	0	275	2.64	0.52	curation error (mis-drawn)
B3DB scaffold	naftifine	3	3.2	0	287	4.99	0.40	
B3DB scaffold	ticlopidine hydrochloride	3	3.2	0	264	3.96	0.29	
B3DB scaffold	desloratadine	2	24.9	1	311	4.02	0.29	
B3DB scaffold	diethylcarbamazine citrate	2	26.8	0	199	0.70	0.35	
B3DB scaffold	rifampin	2	33.2	0	288	3.20	0.28	curation error (name/structure)
B3DB scaffold	B3DB:13	2	36.4	2	161	1.06	0.45	
B3DB scaffold	dapiprazole	5	37.2	0	325	2.29	0.41	
B3DB scaffold	B3DB:2805	3	39.1	1	255	2.53	0.27	
B3DB scaffold	phenylbutazone	2	40.6	0	294	3.40	0.44	curation error (mis-drawn)
B3DB scaffold	dropropizine	2	46.9	2	236	0.16	0.39	
B3DB scaffold	acetohydroxamic acid	2	49.3	2	75	-0.49	0.43	
B3DB scaffold	emetine hydrochloride	2	52.2	1	481	4.94	0.46	
B3DB scaffold	diazoxide	4	58.5	1	231	1.68	0.32	
BBBP random	Phenylbutazone	3	40.6	0	308	3.79	0.35	B3DB: BBB+
BBBP random	Thiopental	6	58.2	2	242	1.35	0.73	B3DB: BBB+
BBBP scaffold	Phenylbutazone	2	40.6	0	308	3.79	0.35	B3DB: BBB+

Table S17: Descriptor contrasts, persistent versus other eligible BBB− molecules (unique molecules). AUC: probability that a persistent molecule has the larger value. Holm over the four primary descriptors and the settings tested within each definition; tests require at least five molecules per group. Support is descriptive.

Definition	Setting	Descriptor	n pers./rest	median pers.	median rest	AUC	Holm
strict	B3DB random	TPSA	10 / 1139	34.09	92.79	0.09	<0.001
strict	B3DB random	HBD	10 / 1139	1.00	2.00	0.19	0.002
strict	B3DB random	MW	10 / 1139	290.88	380.49	0.28	0.044
strict	B3DB random	cLogP	10 / 1139	2.59	2.12	0.62	0.352
strict	B3DB random	support	10 / 1139	0.41	0.53	0.32	–
strict	B3DB scaffold	TPSA	14 / 1185	36.81	93.06	0.09	<0.001
strict	B3DB scaffold	HBD	14 / 1185	0.50	2.00	0.19	<0.001
strict	B3DB scaffold	MW	14 / 1185	269.59	384.65	0.20	<0.001
strict	B3DB scaffold	cLogP	14 / 1185	2.59	2.12	0.56	0.444
strict	B3DB scaffold	support	14 / 1185	0.39	0.44	0.37	–
strict	BBBP random	TPSA	2 / 366	49.41	111.92	–	–
strict	BBBP random	HBD	2 / 366	1.00	3.00	–	–
strict	BBBP random	MW	2 / 366	275.36	400.50	–	–
strict	BBBP random	cLogP	2 / 366	2.57	1.22	–	–
strict	BBBP random	support	2 / 366	0.54	0.59	–	–
strict	BBBP scaffold	TPSA	1 / 385	40.62	110.43	–	–
strict	BBBP scaffold	HBD	1 / 385	0.00	3.00	–	–
strict	BBBP scaffold	MW	1 / 385	308.38	403.83	–	–
strict	BBBP scaffold	cLogP	1 / 385	3.79	1.37	–	–
strict	BBBP scaffold	support	1 / 385	0.35	0.46	–	–
strict	BBBP→B3DB	TPSA	0 / 1070	–	87.34	–	–
strict	BBBP→B3DB	HBD	0 / 1070	–	2.00	–	–
strict	BBBP→B3DB	MW	0 / 1070	–	366.44	–	–
strict	BBBP→B3DB	cLogP	0 / 1070	–	2.47	–	–
strict	BBBP→B3DB	support	0 / 1070	–	0.32	–	–
majority	B3DB random	TPSA	42 / 1107	36.78	94.83	0.14	<0.001
majority	B3DB random	HBD	42 / 1107	1.00	2.00	0.22	<0.001
majority	B3DB random	MW	42 / 1107	287.92	383.52	0.26	<0.001
majority	B3DB random	cLogP	42 / 1107	2.65	2.11	0.58	0.286
majority	B3DB random	support	42 / 1107	0.37	0.54	0.33	–
majority	B3DB scaffold	TPSA	55 / 1144	36.42	95.94	0.13	<0.001
majority	B3DB scaffold	HBD	55 / 1144	1.00	2.00	0.21	<0.001
majority	B3DB scaffold	MW	55 / 1144	288.44	389.64	0.26	<0.001
majority	B3DB scaffold	cLogP	55 / 1144	2.92	2.11	0.60	0.083
majority	B3DB scaffold	support	55 / 1144	0.32	0.44	0.30	–
majority	BBBP random	TPSA	14 / 354	69.28	112.73	0.24	0.010
majority	BBBP random	HBD	14 / 354	2.00	3.00	0.27	0.027
majority	BBBP random	MW	14 / 354	313.12	404.63	0.37	0.286
majority	BBBP random	cLogP	14 / 354	3.04	1.11	0.73	0.027
majority	BBBP random	support	14 / 354	0.44	0.59	0.38	–
majority	BBBP scaffold	TPSA	12 / 374	63.89	111.90	0.22	0.010
majority	BBBP scaffold	HBD	12 / 374	2.00	3.00	0.22	0.009
majority	BBBP scaffold	MW	12 / 374	359.32	405.49	0.35	0.286
majority	BBBP scaffold	cLogP	12 / 374	2.87	1.33	0.70	0.092
majority	BBBP scaffold	support	12 / 374	0.30	0.46	0.26	–
majority	BBBP→B3DB	TPSA	46 / 1024	36.00	89.06	0.19	<0.001
majority	BBBP→B3DB	HBD	46 / 1024	1.00	2.00	0.26	<0.001
majority	BBBP→B3DB	MW	46 / 1024	287.92	371.32	0.31	<0.001
majority	BBBP→B3DB	cLogP	46 / 1024	2.80	2.44	0.56	0.286
majority	BBBP→B3DB	support	46 / 1024	0.31	0.32	0.49	–

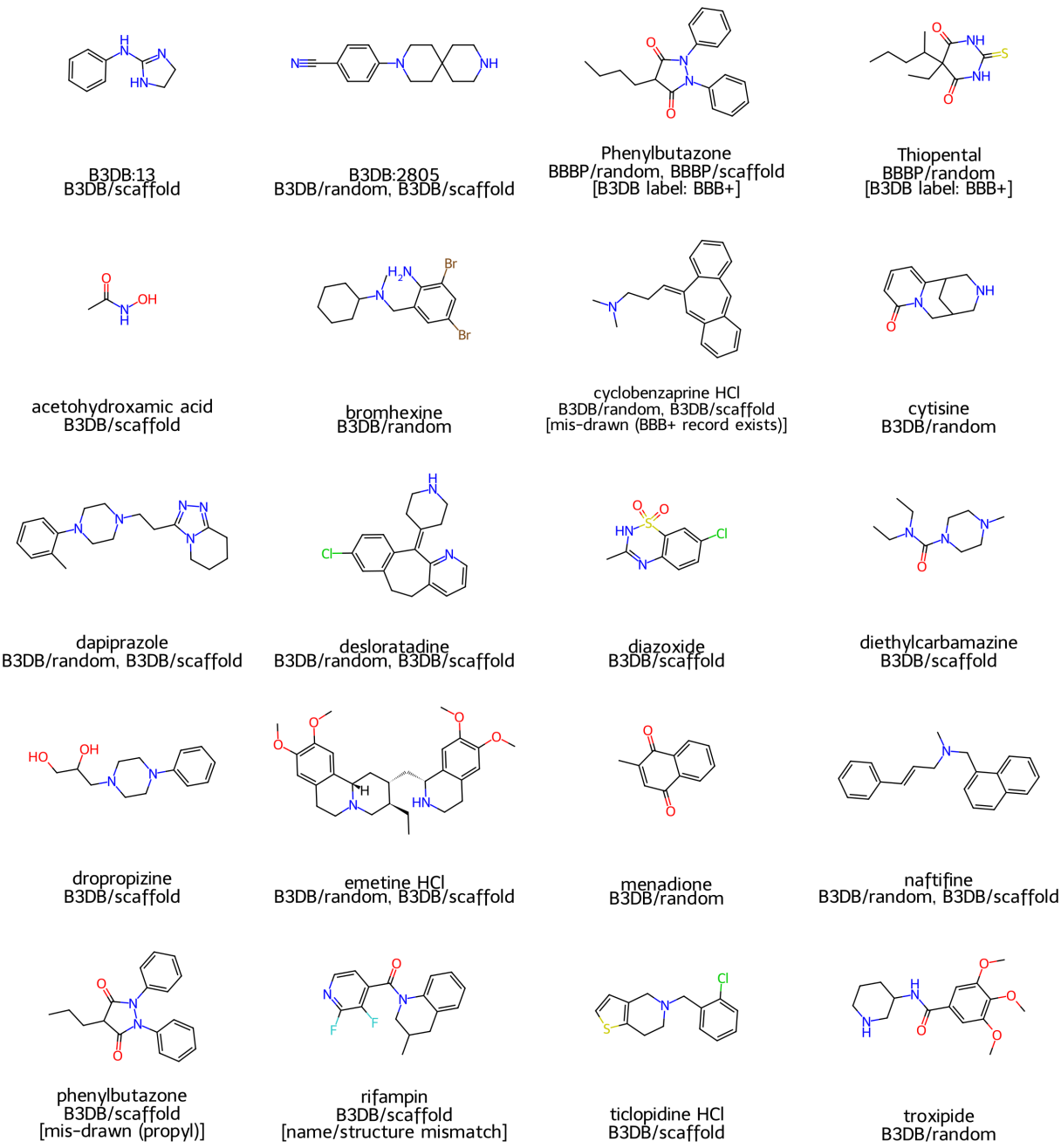


Figure S7: Structures of cross-architecture persistent BBB− failures (unique molecules, strict definition), with the settings in which each is persistent and dataset flags.

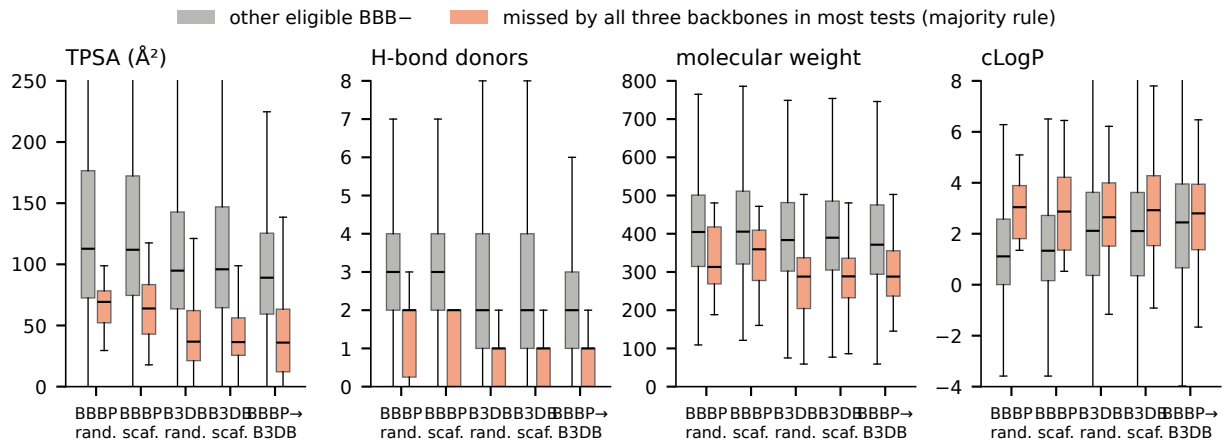


Figure S8: Descriptors of shared failures (majority rule, which populates all five settings) against other eligible BBB− molecules; unique molecules, boxes show quartiles. Holm-adjusted tests are in Table S17.

Table S18: TPSA association with being missed, adjusted for support. Logistic regression of a miss under class-conditional calibration on standardized descriptors over all BBB− test appearances; coefficient and 95% interval from 400 bootstrap resamples of splits. Permutation p: TPSA permuted within support tertiles (500 permutations).

Model	Setting	TPSA, unadjusted	+ support	+ support, MW, cLogP	permutation p
RF	BBBP random	-3.07 [-3.56, -2.67]	-2.85 [-3.32, -2.46]	-2.78 [-3.89, -1.83]	<0.002
RF	BBBP scaffold	-2.86 [-3.90, -1.92]	-2.44 [-3.52, -1.33]	-0.80 [-2.19, 0.41]	<0.002
RF	B3DB random	-2.12 [-2.69, -1.63]	-2.12 [-2.67, -1.63]	-2.23 [-2.98, -1.41]	<0.002
RF	B3DB scaffold	-2.11 [-2.59, -1.62]	-2.04 [-2.50, -1.58]	-2.90 [-3.55, -2.21]	<0.002
GIN	BBBP random	-1.54 [-1.98, -1.14]	-1.36 [-1.86, -0.97]	-3.00 [-3.83, -2.05]	<0.002
GIN	BBBP scaffold	-0.89 [-1.74, -0.20]	-0.46 [-1.20, 0.13]	-1.22 [-2.89, 0.50]	<0.002
GIN	B3DB random	-1.13 [-1.44, -0.87]	-1.03 [-1.32, -0.71]	-1.82 [-2.31, -1.42]	<0.002
GIN	B3DB scaffold	-1.21 [-1.62, -0.80]	-1.11 [-1.54, -0.72]	-2.76 [-3.34, -2.24]	<0.002
Chemprop	BBBP random	-1.73 [-2.19, -1.30]	-1.49 [-2.02, -1.04]	-2.65 [-3.83, -1.61]	<0.002
Chemprop	BBBP scaffold	-2.01 [-2.86, -1.41]	-1.63 [-2.56, -0.96]	-0.85 [-1.77, 0.11]	<0.002
Chemprop	B3DB random	-1.61 [-1.98, -1.27]	-1.50 [-1.84, -1.15]	-1.91 [-2.57, -1.27]	<0.002
Chemprop	B3DB scaffold	-1.57 [-1.97, -1.14]	-1.43 [-1.83, -1.04]	-2.44 [-3.00, -1.76]	<0.002

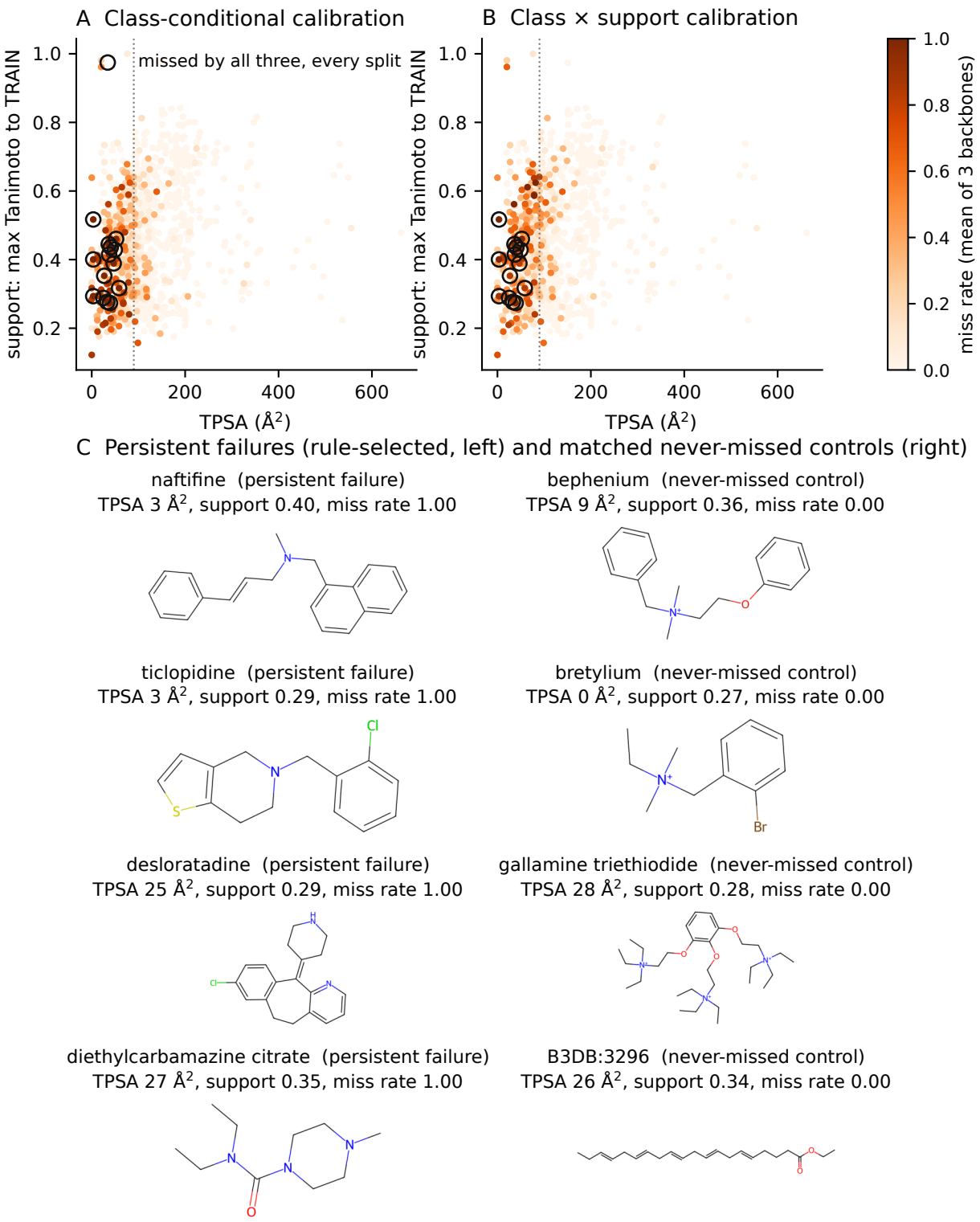


Figure S9: Chemistry of the misses, B3DB scaffold splits. (A, B) Every eligible BBB− test molecule by TPSA and mean support, coloured by its miss rate (mean over the three backbones) under class-conditional (A) and class × support (B) calibration; circles: strict cross-architecture failures. (C) Strict failures selected by a fixed rule (lowest TPSA first, the three curation-error records excluded), each with the never-missed BBB− molecule nearest in TPSA and support. Several controls are permanently charged; we did not test this pattern on independent data and report it only as a hypothesis.

G Screening decisions

Table S19: Decision metrics at $\alpha = 0.1$, mean over three backbones and 20 splits. Rejection baselines are matched per split to the deferral rate of class × support (C×S); in the VAL-threshold variants the deferral rate and the rejection thresholds are set on VAL, so they are not matched on the test set.

Setting	Rule	deferral	accepted error	BBB− accepted as BBB+	low-support accepted error
BBBP random	marginal	0.067	0.101	0.286	0.147
	Mondrian	0.210	0.128	0.089	0.199
	C×S	0.306	0.125	0.069	0.225
	AD rejection	0.306	0.102	0.194	0.150
	conf. rejection	0.306	0.057	0.126	0.094
	AD rejection, VAL threshold	0.310	0.101	0.192	0.157
	conf. rejection, VAL threshold	0.322	0.055	0.120	0.090
BBBP scaffold	marginal	0.139	0.115	0.322	0.167
	Mondrian	0.311	0.160	0.095	0.241
	C×S	0.383	0.155	0.076	0.196
	AD rejection	0.383	0.124	0.216	0.099
	conf. rejection	0.383	0.070	0.146	0.113
	AD rejection, VAL threshold	0.421	0.122	0.196	0.085
	conf. rejection, VAL threshold	0.409	0.067	0.139	0.109
B3DB random	marginal	0.270	0.131	0.194	0.209
	Mondrian	0.303	0.137	0.094	0.210
	C×S	0.322	0.134	0.086	0.185
	AD rejection	0.322	0.172	0.205	0.232
	conf. rejection	0.322	0.116	0.160	0.194
	AD rejection, VAL threshold	0.325	0.172	0.205	0.240
	conf. rejection, VAL threshold	0.323	0.116	0.160	0.193
B3DB scaffold	marginal	0.291	0.145	0.203	0.234
	Mondrian	0.317	0.152	0.110	0.243
	C×S	0.340	0.150	0.102	0.221
	AD rejection	0.340	0.185	0.208	0.249
	conf. rejection	0.340	0.132	0.174	0.220
	AD rejection, VAL threshold	0.357	0.184	0.202	0.222
	conf. rejection, VAL threshold	0.345	0.130	0.173	0.218
BBBP→B3DB	marginal	0.234	0.378	0.532	0.454
	Mondrian	0.378	0.319	0.186	0.393
	C×S	0.495	0.297	0.127	0.371
	AD rejection	0.495	0.339	0.295	0.421
	conf. rejection	0.495	0.336	0.321	0.426
	AD rejection, VAL threshold	0.647	0.298	0.179	0.401
	conf. rejection, VAL threshold	0.599	0.317	0.244	0.416

Table S20: Class-conditional calibration plus a rejection filter at $\alpha = 0.1$, mean over three backbones and 20 splits. The filter defers, among the singleton calls of class-conditional calibration, the compounds with the lowest support (AD) or confidence until total deferral equals that of class × support on the same split (matched without test labels); VAL rows set the extra deferral and the cut-off on VAL. Where class-conditional calibration already defers more, no extra deferral is added.

Setting	Rule	deferral	accepted error	BBB− accepted as BBB+
BBBP random	C×S	0.306	0.125	0.069
	Mondrian	0.210	0.128	0.089
	Mondrian + AD filter	0.312	0.113	0.062
	Mondrian + conf. filter	0.312	0.085	0.083
	Mondrian + AD filter, VAL	0.309	0.112	0.062
	Mondrian + conf. filter, VAL	0.315	0.083	0.082
BBBP scaffold	C×S	0.383	0.155	0.076
	Mondrian	0.311	0.160	0.095
	Mondrian + AD filter	0.394	0.145	0.073
	Mondrian + conf. filter	0.394	0.124	0.089
	Mondrian + AD filter, VAL	0.410	0.142	0.066
	Mondrian + conf. filter, VAL	0.412	0.115	0.089
B3DB random	C×S	0.322	0.134	0.086
	Mondrian	0.303	0.137	0.094
	Mondrian + AD filter	0.326	0.132	0.085
	Mondrian + conf. filter	0.326	0.128	0.094
	Mondrian + AD filter, VAL	0.324	0.132	0.085
	Mondrian + conf. filter, VAL	0.327	0.128	0.094
B3DB scaffold	C×S	0.340	0.150	0.102
	Mondrian	0.317	0.152	0.110
	Mondrian + AD filter	0.344	0.146	0.099
	Mondrian + conf. filter	0.344	0.142	0.110
	Mondrian + AD filter, VAL	0.347	0.145	0.098
	Mondrian + conf. filter, VAL	0.341	0.144	0.110
BBBP→B3DB	C×S	0.495	0.297	0.127
	Mondrian	0.378	0.319	0.186
	Mondrian + AD filter	0.497	0.295	0.133
	Mondrian + conf. filter	0.497	0.311	0.179
	Mondrian + AD filter, VAL	0.563	0.274	0.111
	Mondrian + conf. filter, VAL	0.553	0.305	0.172

H Verification and reproducibility

Each number in the main text is a macro written by `scripts/make_paper_numbers.py` together with its source file, filter and aggregation (`generated/numbers_provenance.csv`). `scripts/verify_paper_numbers.py`, which imports only NumPy, pandas, SciPy and RDKit and shares no code with the analysis pipeline, recomputes each value from the cached predictions with its own conformal quantile, stratum assignment, count-based merging rule, Wilcoxon and Holm procedures and logistic fit, and records whether each value was recomputed exactly, replicated with independent random draws within a stated tolerance, or cross-checked against per-split outputs. Persistent-failure results are recomputed by `scripts/verify_persistent_failures.py`. The analysis reproduces the per-split coverages of the evaluation pipeline exactly before any per-molecule result is computed. Exploratory results that depend on the pipeline's own randomization (random strata) or on implementations of published methods are cross-checked against per-split outputs rather than recomputed. Counts that depend on whether a bootstrap interval excludes zero are replicated with independent draws and may differ by the number of cells whose interval endpoint lies within 0.02 of zero; such counts are not used as firm results. In a few small splits a support tertile contains no BBB− test molecule; that split contributes no value for the tertile, identically in the analysis and in the verifier. All analyses of the cached predictions were reproduced on a second computing platform with identical results; models retrained there for the additional tasks gave identical splits but slightly different predictions (a different

RDKit release), with unchanged conclusions. Package versions are pinned in `requirements-lock.txt`. All code, data and commands are available in the public repository listed in the main text.

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